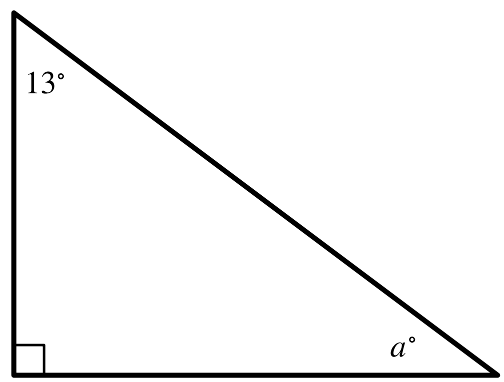
Question ID: 69f4bbdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of a?

Answer

A. 13

B. 77

C. 90

D. 103

Correct Answer: B

Rationale

Choice B is correct. The triangle shown is a right triangle, where the interior angle shown with a right angle symbol has a measure of $90°$. It's shown that the other two interior angles measure $13°$ and $a°$. The sum of the measures of the interior angles of a triangle is $180°$; therefore, $90 + 13 + a = 180$. Combining like terms on the left-hand side of this equation yields $103 + a = 180$. Subtracting 103 from both sides of this equation yields $a = 77$.

Choice A is incorrect. This is the measure, in degrees, of the other acute interior angle of the right triangle, not the value of a.

Choice C is incorrect. This is the measure, in degrees, of the right angle of the right triangle, not the value of a.

Choice D is incorrect. This is the sum of the measures, in degrees, of the other two interior angles of the right triangle, not the value of a.